Università degli Studi di Padova

Dipartimento di Ingegneria dell'Informazione




# Digital Forensics and Biometrics

A.A. 2026/2027

## Introduction to the course

*Simone Milani*
*Room 216*
*DEI A*

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*E-mail:*
*simone.milani@dei.unipd.it*

# INFORMATION ABOUT THE COURSE



Simone Milani
Dept. of Information Engineering Room 216 - DEI A – 2° floor
phone: 049 827 7641 e-mail: simone.milani@dei.unipd.it
Home page: https://medialab.dei.unipd.it/members/simone-milani/

The time schedule of the course is:

- Tue 12:30-14:30 room 1C150 TORRE ARCHIMEDE
- Fri 8:30-10:30 room D COMPLESSO PIOVEGO

Meeting hours with teacher: **Tuesdays 9:30-10:30 with preliminary booking (send an email).**



Teaching assistant:
Mattia Tamiazzo (mattia.tamiazzo@student.unipd.it)

# LAB SESSIONS

During the semester, we are planning 5 lab sessions in room Lab TA (Torre Archimede).

Experience will be based on Python using Google CoLab.

Make sure you have an account on Lab TA PCs. You can use your personal laptop, if you desire.

Scheduled dates are

- Oct. 20 @ 12:30
- Nov. 3 @ 12:30
- Nov. 17 @ 12:30
- Dec. 1 @ 12:30
- Dec. 15 @ 12:30



# EXAMINATION

The evaluation will be based on a **written exam** on the theoretical part and a **final project** XOR **legal report**. Total points: 32 (max).

Written exam will be made of 3 open questions/problems + 30 multiple choice questions.

Dates are reported on Uniweb.

It is possible to anticipate the written exam with two pre-term exams (Dec. 18 + Jan. 12)

**Theoretical written**
(intermediate or normal sessions)



+

**Project**



Project/legal reports are to be described with a 10 minutes presentation. See presentation dates on Uniweb.

XOR

**Legal report**



Legal report on a digital evidence

# TENTATIVE EXAM DATES

**On Uniweb!! All variations and times shifts will be notified on Uniweb!!!**

**Written exam e report dates**


Jan 21, 2027 @ 9:30


Feb 11, 2027 @ 9:30


July 08, 2027 @ 9:30


Aug. 24, 2027 @ 9:30

**Project presentation dates**


Feb. 11, 2027 @ 14:30


Aug. 24, 2027 @ 14:30

Usually, for project presentations a calendar of the time slots will be provided. Report do not need to be presented (but uploaded by the dealine reported on Moodle page)

**PLEASE enroll on Uniweb** under the section "Partial exams".

**You need to have DFB in your study plan!**

# INTERMEDIATE EXAMS

In place of the final theoretical exam, you can **take two intermediate exams**. Average score will be the score for the theoretical part.



- Dec. 18, 2026 time 12:30 room P140
- Jan. 12, 2027. time 12:30 room P140

Intermediate exam lasts 1 hour and will cover half program. 15 M-choice questions + 2 open questions

You can access the second intermediate exam if your score is **higher or equal to 16/30** in the first intermediate exam.

**Fastest track:** two intermediate + project/report on Jan 21 (depending on requests).

# EXAM GRADING: WRITTEN

**Theoretical part:**
Every question will contribute to the final score with a predefined number of points.

Multiple choice score will be mapped to points with arithmetic rescaling function.

Open questions will be evaluating taking into account:

| Correctness | Completeness | Justifications | Details | Clarity | Order |
|---|---|---|---|---|---|

The resulting written score will range from 0-31.

Such evaluation will be mapped in the range [0-26] or [0-28] depending on what you choose as practical proof (project or report).

# EXAM GRADING: PROJECT OR REPORT

## Final project (0-6 points)

You can form groups of **2 person at max**. per project. It is possible to develop a project in different languages but Python is preferred. You must communicate project preferences and a title will be assigned to you.

Students will need to present the project before the exam date by **preparing a presentation and submitting the source code.** Evaluation will be performed considering:

| Originality | Accuracy | Completeness | Details | Presentation | Clarity |
|---|---|---|---|---|---|

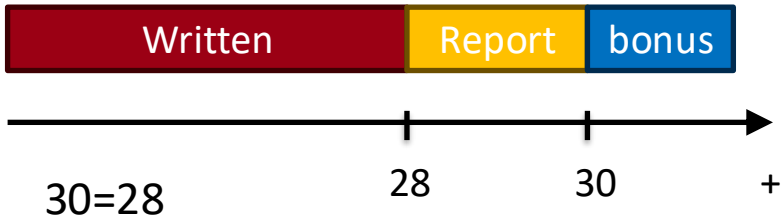
Two project presentation date (more could be arranged if necessary).
Score can change for each team member depending on answers**: members contributions must be clearly reported in the project materials/presentation**!

## Final report (0-2 points)

Legal report are focused on the analysis of a **digital evidence** that are provided to you by the instructor. You must ask book an evidence in advance via the form on the moodle page of the course.
Legal report must be uploaded as compressed folder **before the report evaluation deadline** including all the materials. No presentation is needed.
Reports are **individual.**

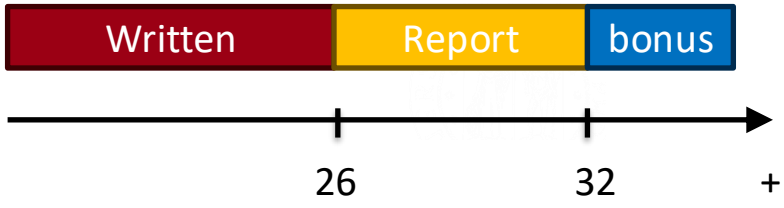
# FINAL SCORE

Total amount of points: 31 (minimum: 18)




You chose **legal report:**

- Written exam accounts for points 0-28: 18=16,8 30=28
- Final report will give you up to 2 points: 0=18. 1=24. 2=30
- Up to 1/2 additional points for attendance




You chose **programming project:**

- Written exam accounts for points 0-26: 18=15,6 30=26
- Final project will give you up to 6 points: 1=18. 6=30L
- Up to 1/2 additional points for attendance

*Examples:*
*Written 28 (mapped to 24,26) Project: 5,3 pts Final score:* ***30 + bonus***
*Written 28 (mapped to 26,1) Report: 2 pts Final score:* ***28 + bonus***
*Written 25 (mapped to 21,67) Project: 5 pts Final score:* ***27 + bonus***
*Written 25 (mapped to 23,33) Report: 1 pts Final score:* ***24 + bonus***
*Written 18 (mapped to 15,6) Project: 5 pts Final score:* ***20 + bonus***
*Written 18 (mapped to 16,8) Report: 2 pts Final score:* ***19 + bonus***
*Written 27 (mapped to 23,4) Project: 30L (6 pts) Final score:* ***29 + bonus***

# MATERIALS

**Slides, notes, and other documents from the instructor**

**Main references:**

Klette, Reinhard, Concise Computer Vision. Springer London: --, 2014.

A. K. Jain, A. A. Ross, K. Nandakumar, "Introduction to Biometrics", Springer 2011

Hany Farid, Photo Forensics, MIT Press, 2019

**ML references:**

Bishop, Christopher M., Pattern recognition and machine learning. New York: Springer,

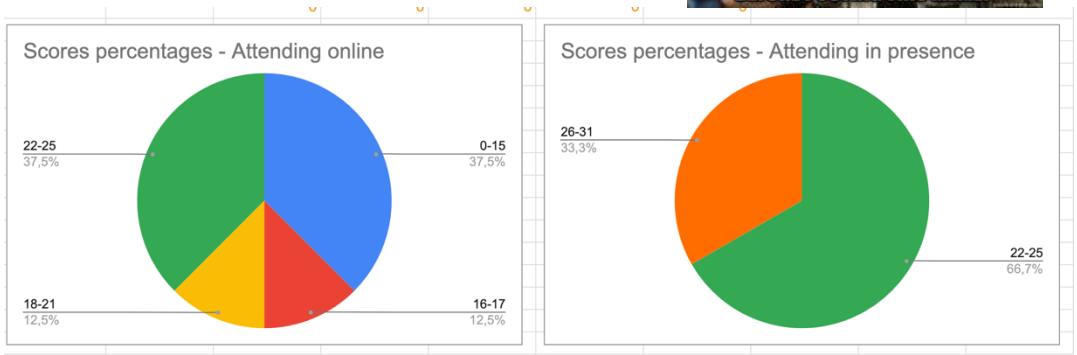
J. Watt, R. Borhani, A. Katsaggelos, "Machine learning refined: foundations, algorithms, and applications" New York: Cambridge University Press, 2016.,

# SOME SUGGESTIONS ...




- Make sure you have checked all the material on the course moodle page before asking questions: most of the information is there!

- Attendance is very important:
  be on the spot in time!




*Data: A.A. 21/22*

- Check exercises/questions and make sure you have understood why an anser is correct or not.

- Check exam correction!

- Better prepare for the exam rather than trying
  (you save time)



# QUESTIONS?

# Introduction to Digital Forensics and Biometrics

# WHAT IS DIGITAL FORENSICS?

From NIST (National Institute of Standards and Technology) Guide to Integrating Forensic Techniques into Incident Response

*"Application of science to the identification, collection, examination, and analysis of data while preserving the integrity of the information and maintaining a strict chain of custody for the data"*

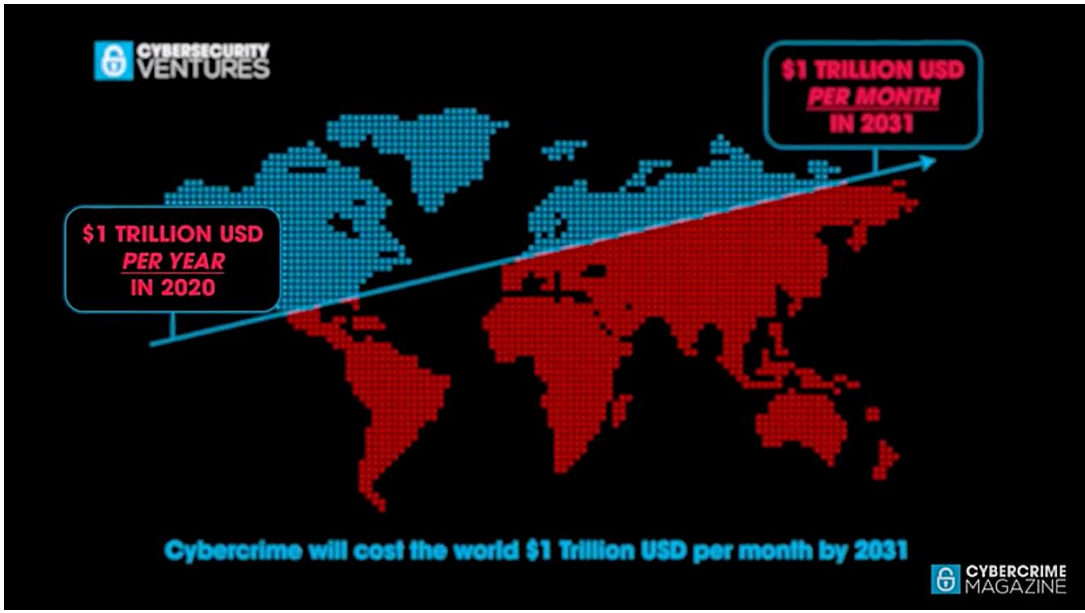


# NEED FOR DIGITAL FORENSICS

No major organization can afford to be without a digital forensics capability due to the amount of data in play and of the importance of that data to the organization.

If that data is damaged, stolen, or otherwise compromised, the organization needs the capability to reconstruct what occurred.

Need aldo to verify and understand problems, security breaches, attacks, failures and malfunctionings.




**Motivations:**

- Predictions say that cybercrime will cost $10.5 trillion annually by the end of 2025 [source: DeepStrike]
- Riskiest industries: Critical infrastructures, banks, telecommunications, technology, chemicals, energy, and transportation services [source: Moody's]

GDP 2026 (expected)



USA 32,49T



China 20,85T



Cybercrime 10.5T



Germany 5,45T



Japan 4,38T

# SECURITY VS. FORENSICS

What is the difference between

Cyber (network, information, etc.) Security and Digital Forensics?



**VS**




**Cyber Security** refer to the set of policies and practices adopted to prevent and monitor unauthorized access, misuse, or modification of digital data, a computing device, a computer network or network-accessible resources.

**Digital Forensics** relates to the monitoring and the analysis of digital evidences (devices, data, network traffic, etc.) with the purpose of reconstructing its story and understand/prove what happened/what has been done.

The latter usually comes into play after the crime/alteration has been done.

# 5 W + H: FORENSIC INVESTIGATIONS

- **Who ?** Who was involved? Suspect, victim, account, device, IP address, …
  Digital ID (e.g., IP address)
  Biological ID (real identity)

  **"Onlife":** life in a world where online and offline are blended

  **Biometrics** (bio=life, metrics=measures)
  *Measurement of physical characteristics or personal traits*

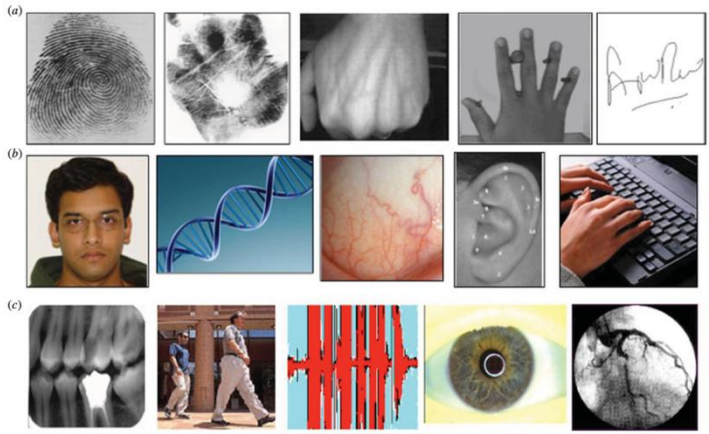
- **What ?** What happened? Data theft, malware infection, image manipulation, …
- **When ?** When did it happen? Timeline, file timestamps, log entries, creation/edit dates.
- **Where ?** Where did it happen? GPS location, device, server, network segment, ….
- **Why ?** Why did it happen? Motivation or objective
- **How ?** How was this done? tools used, exploit, credentials, malware, social engineering, …

To get the facts, not necessarily guilt or innocence



# FORENSIC CALL FOR BIOMETRICS ....

Who or what is the source (person) related to this evidence?



Possible outcome:

- Individualization
- Inconclusive
- Exclusion

Associate an evidence (measure) to an individual but

1. Forensic analysis is applied after the event has occurred; biometrics can be applied before the event (access control).
2. In a forensic investigation you can not control in advance the type of measurement
3. Forensic evidence collection is mostly manual
4. Biometric recognition probably needs to be done in real time
5. In forensics, false negative is to be minimize: exclude the perpetrator. In biometrics, it depends: in surveillance, false non-matches need to be minimize w.r.t. false matches; in access control, avoid false matches
6. Inconclusive can be compensated in biometrics by additional acquisition
7. In forensics, measures have a lower quality
8. The outcome of a forensic investigation has to be declared verbally to a jury or a court; in biometrics, outcomeis parameterized numerically

# SOME EXAMPLES

- Allianz Life Jul. 26, 2025
  *social engineering to access a third-party cloud system and obtain personal data link*

- SHAMS II operation by Interpol link . Sep 18, 2026
  extracted more than 100k facial images from visual content published by jihadist groups, filtered them, and processed them through INTERPOL's facial-recognition system

- Clear view case Sep. 3, 2024
  Clearview built a large facial-recognition database from images scraped online. Forensic and legal analysis focused on how images were collected,link

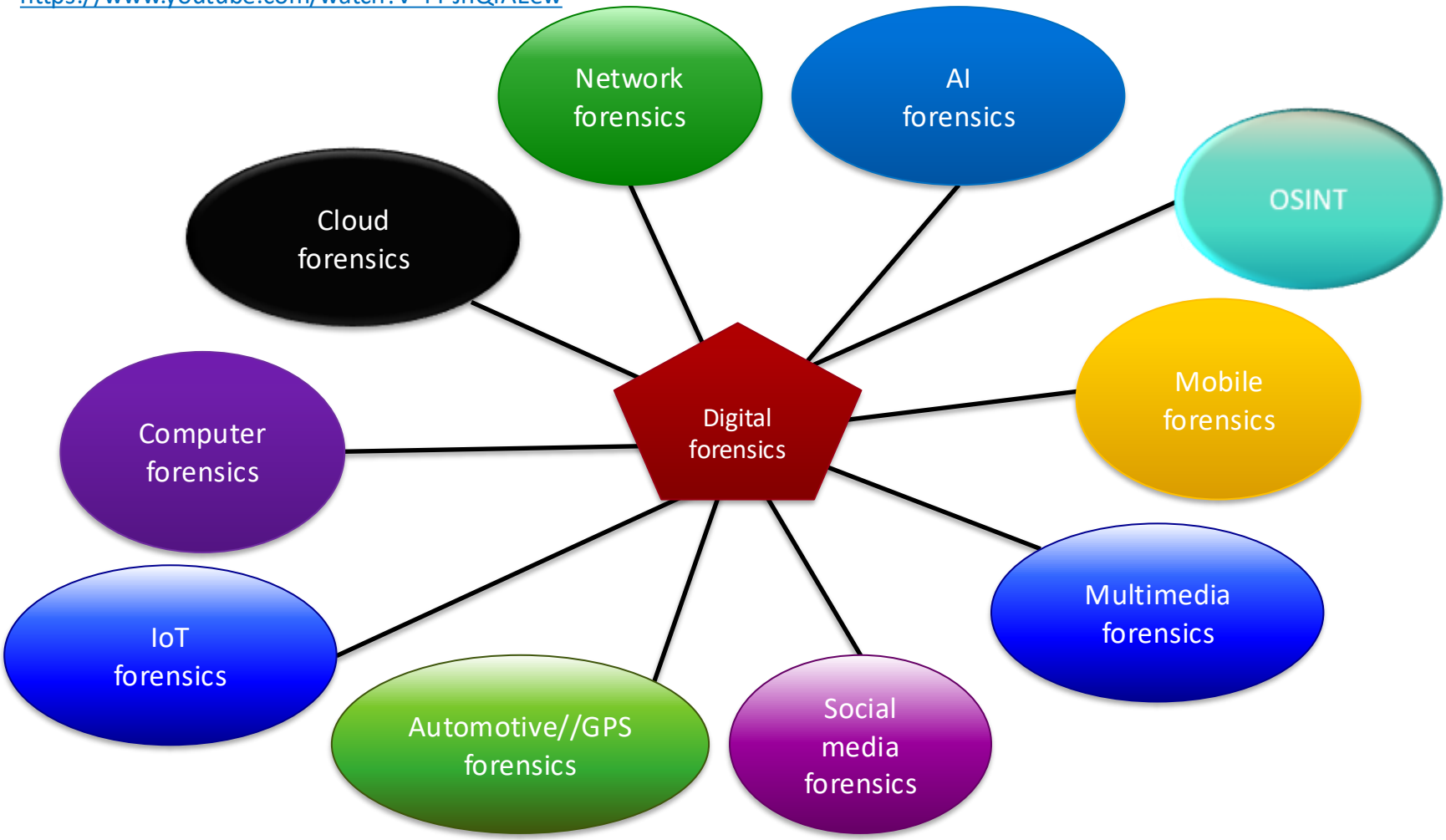
- Anomaly detection for industrial production and safety



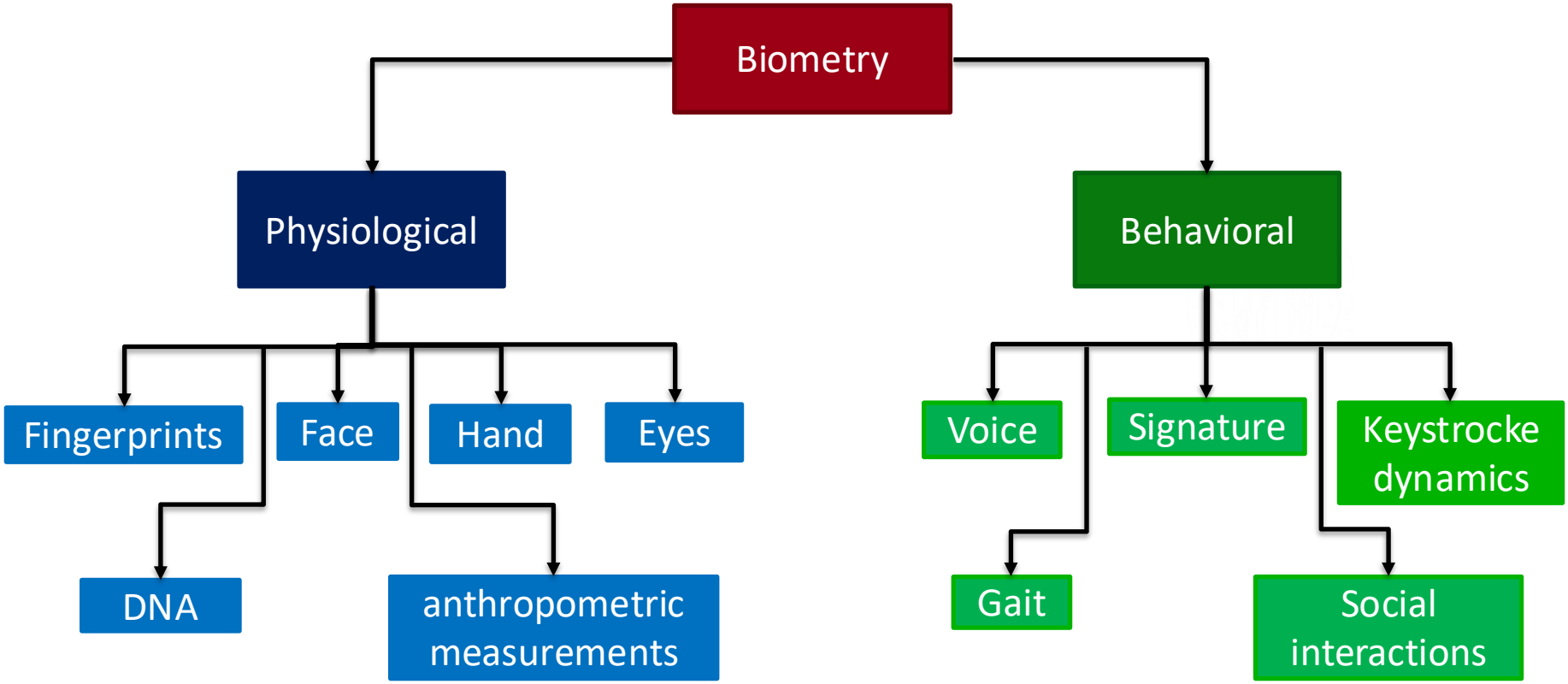
# DIGITAL EVIDENCES

Current investigations and analysis are characterized by the use of digital evidences, i.e., digital devices or data (images, files, network streams, etc.)

Digital forensics take care of this!
https://www.youtube.com/watch?v=Pf-JnQfAEew



# FROM DIGITAL EVIDENCE TO REAL LIFE ID



# WHAT WILL YOU GAIN IN THIS COURSE?

- Knowledge of the main analysis strategies for disk, network, multimedia and social media forensics
- Knowledge of the main biometric strategies and traces that can be exploited for people identification.
- Ability to perform a forensics investigation and write a report
- Knowledge of the main legal issues related to cyber crimes
- How machine and deep learning can be used to carry on an investigation.

## Roadmap



# ACQUIRING A DIGITAL EVIDENCE

# DIGITAL EVIDENCES



# ACQUIRING AND PRESERVING DIGITAL EVIDENCES

## *ISO IEC 27037/2012*

International standard defining guidelines for the identification, acquisizition and the preservation of digital evidences.



Security in handling information

ISO: International Organization for Standardization

- Most widely-known organization for the definition of technical guidelines
- Founded feb. 23, 1047 – headquarter: Geneva, Switzerland
- 162 countries participating
- cooperates with IEC

IEC: International Electrotechnical Commision

- Standardization of electrical, electronic and related technologies
- Founded 1906; headquarter: Geneva, Switzerland
- 60 countries cooperating
- Commision: representatives of acknowleged standardization organization

# OTHER STANDARDS

**ISO/TR 15801:2009**
Document management - Information stored electronically –
Recommendations for trustworthiness and reliability

**ISO/IEC 17020:2012**
Conformity assessment - Requirements for the operation of various types of bodies performing inspection

**ISO/IEC 17025:2005**
General requirements for the competence of testing and calibration laboratories

**ISO/IEC 27000:2012**
Information technology - Security techniques – Information security management systems - Overview and vocabulary

**ISO/IEC 27041**
Information technology – Security techniques – guidance for ensuring that methods and processes used in the investigation are "fit for purpose".

**ISO/IEC 27042**
Information technology – Security techniques – guidance on the analysis and interpretation of digital evidences w.r.t. continuity, validity, reproducibility, and repeatability.

**ISO/IEC 27043**
Information technology – Security techniques – guidelines based on idealized models for common incident investigation processes.

# ISO IEC 27037/2012

In a nutshell: defines how to process digital data in a legal issue.

**Note:** data are considered already in digital format (do not consider analog data converted into digital one)!

Specifies:

- Identification (inspection)



- Seizure (virtual)



- Acquisition

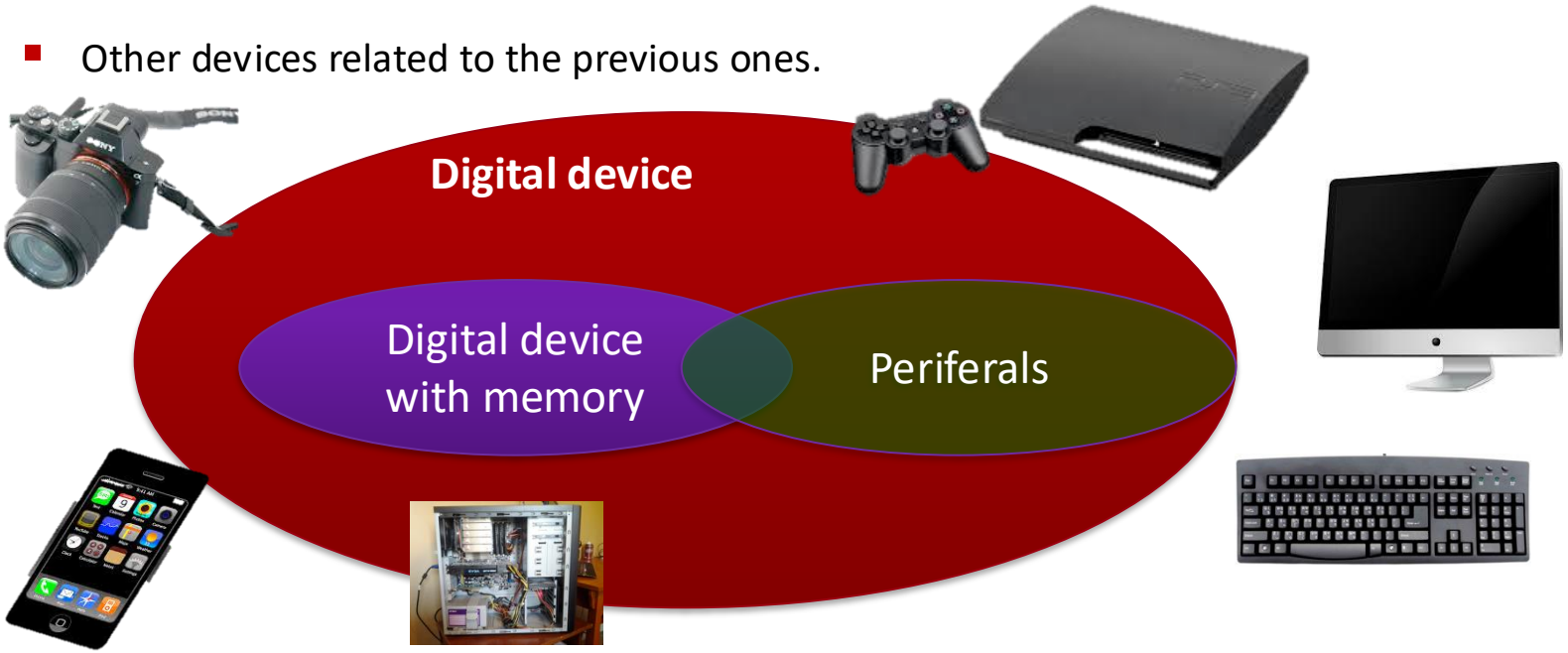


- Preservation (conservation and sealing)



# ISO IEC 27037/2012: MEMORY DEVICES

- Memory devices inside computers (hard disks, floppy disks, optical disks, magentic/optical supports, USB pens, SD cards, etc.)
- Cellphones, Personal Digital Assistants (PDAs), Personal Electronic Devices (PDEs), memory cards, GPS and navigation systems.
- Photocameras and videocameras (including CCTVs)
- Computer with network connections
- Network devices based on TCP/IP or other protocols
- Other devices related to the previous ones.



# REQUIREMENTS

- **Pertinence**
  - DE must be useful to prove guilty or innocent
  - Documentation must prove that the DE is relevant of worth of acquisition

- **Reliability**
  - Ensure that the evidence is authentic (no alterations)
  - All the processes and operations applied on the DE must be documented and reproducible.

- **Adequateness**
  - DEFR must evaluate which/how much evidences must be acquired (and the methodologies to be adopted)
  - Material can be copied or seized
  - A complete copy is not always required
    - Depends on the case
    - And on the national law systems

- **Verificability**
  - A third party needs to verify the activities of DEFR and DES
    - Need for documentation that allows this
    - Scientific method, techniques and procedures must be evaluated as well
  - DEFR and DES must be able to justify their operations

- **Repeatability, Reproducibility**
  - Operations can be repeated using the same method, procedures, tools, and conditions
  - Results can be reproduced using the same method, but different tools and conditions

- **Provable**
  - Prove that the adopted methods are the best possible

# ISO IEC 27037/2012: WHAT AND WHAT NOT?

- Logging (keep documentation of each phase of the analysis)
- Chain of custody (traceability)
- Priorities (plan the investigation activity)
- Protection (sealing and packaging of the devices)
- Transportation of evidences (real/virtual)
- Roles in evidence exchanges (who & why)

But you won't find …

- Legal issues (international)
- Analysis
- Technical instruments
- Report writing and presentation
- Dealing with analogic data